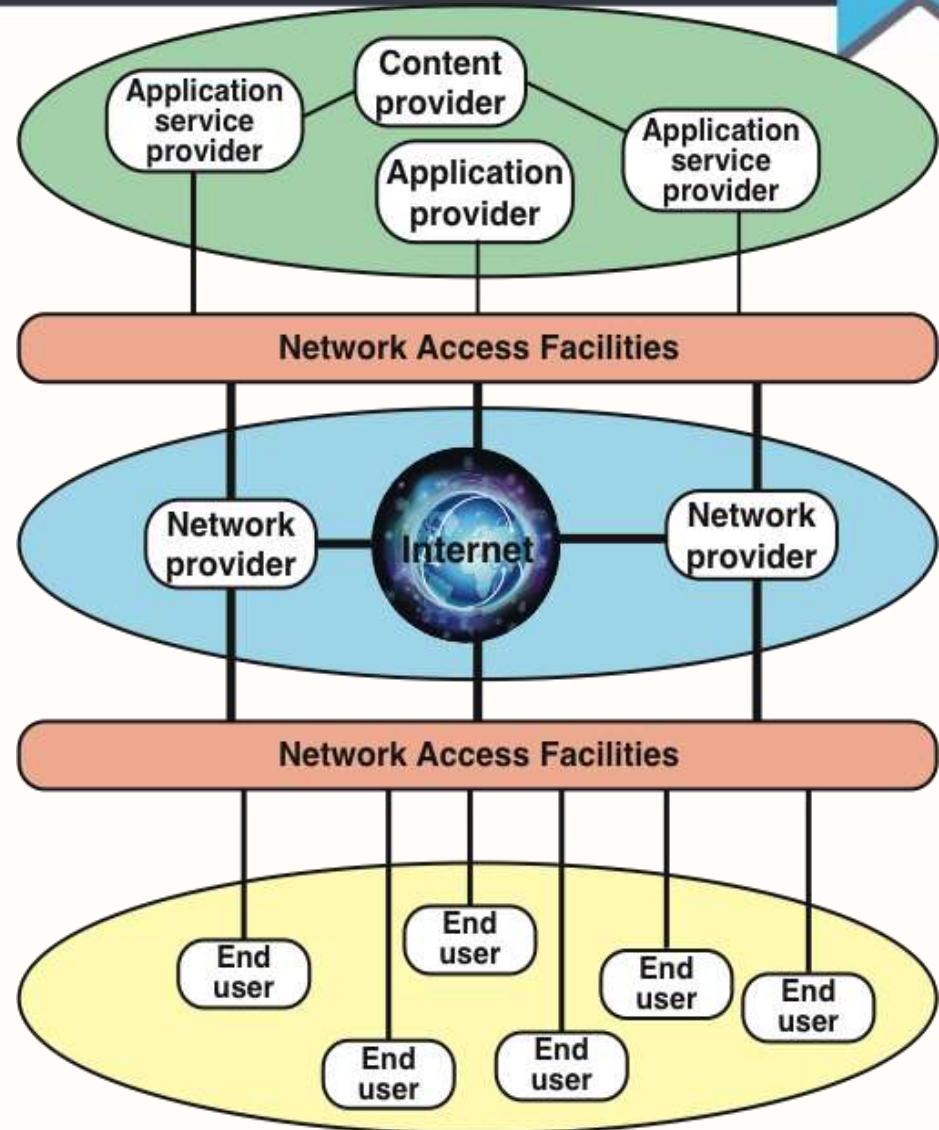


Software Defined Networking

6CS029 Advanced Networks

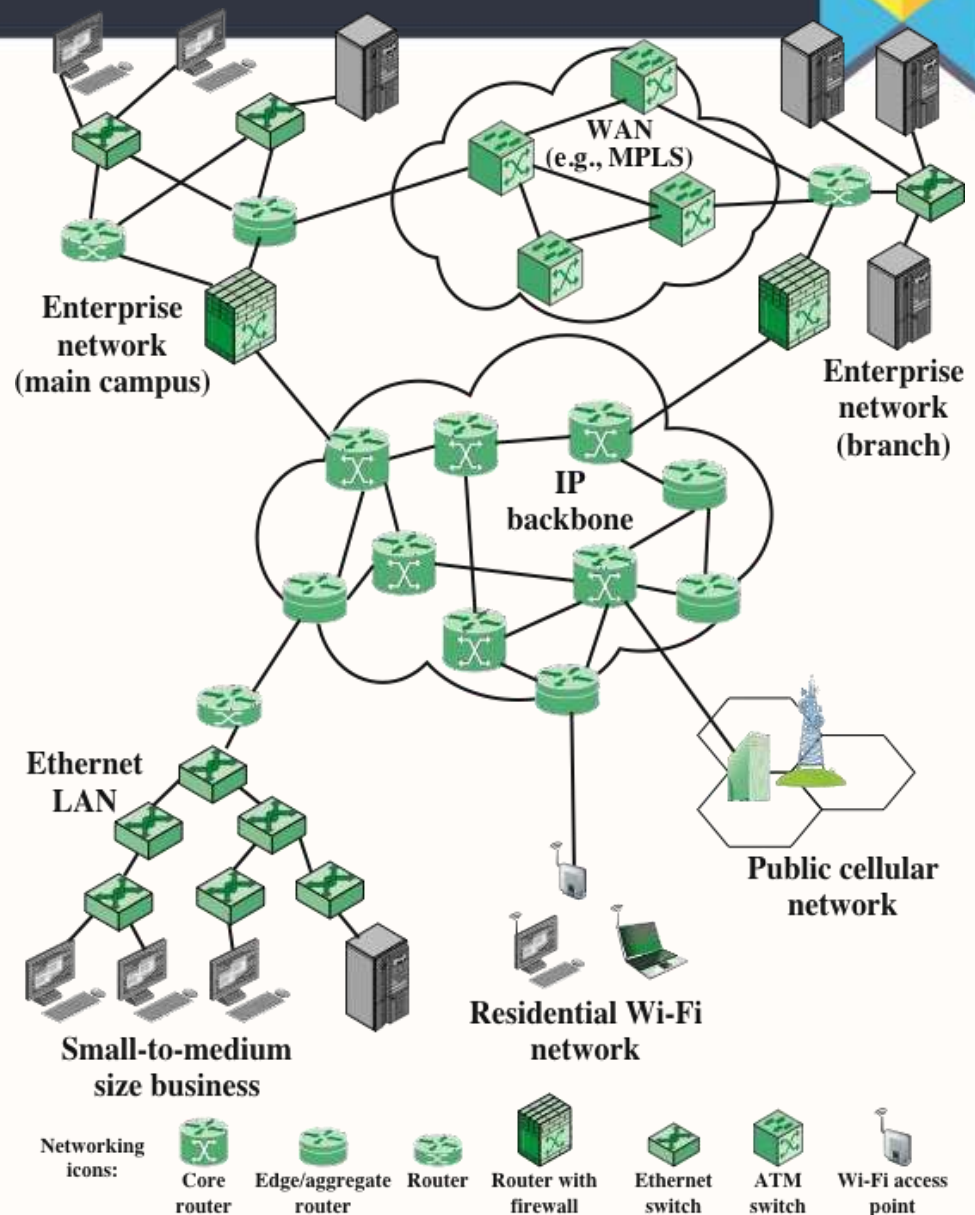


The Modern Networking Ecosystem



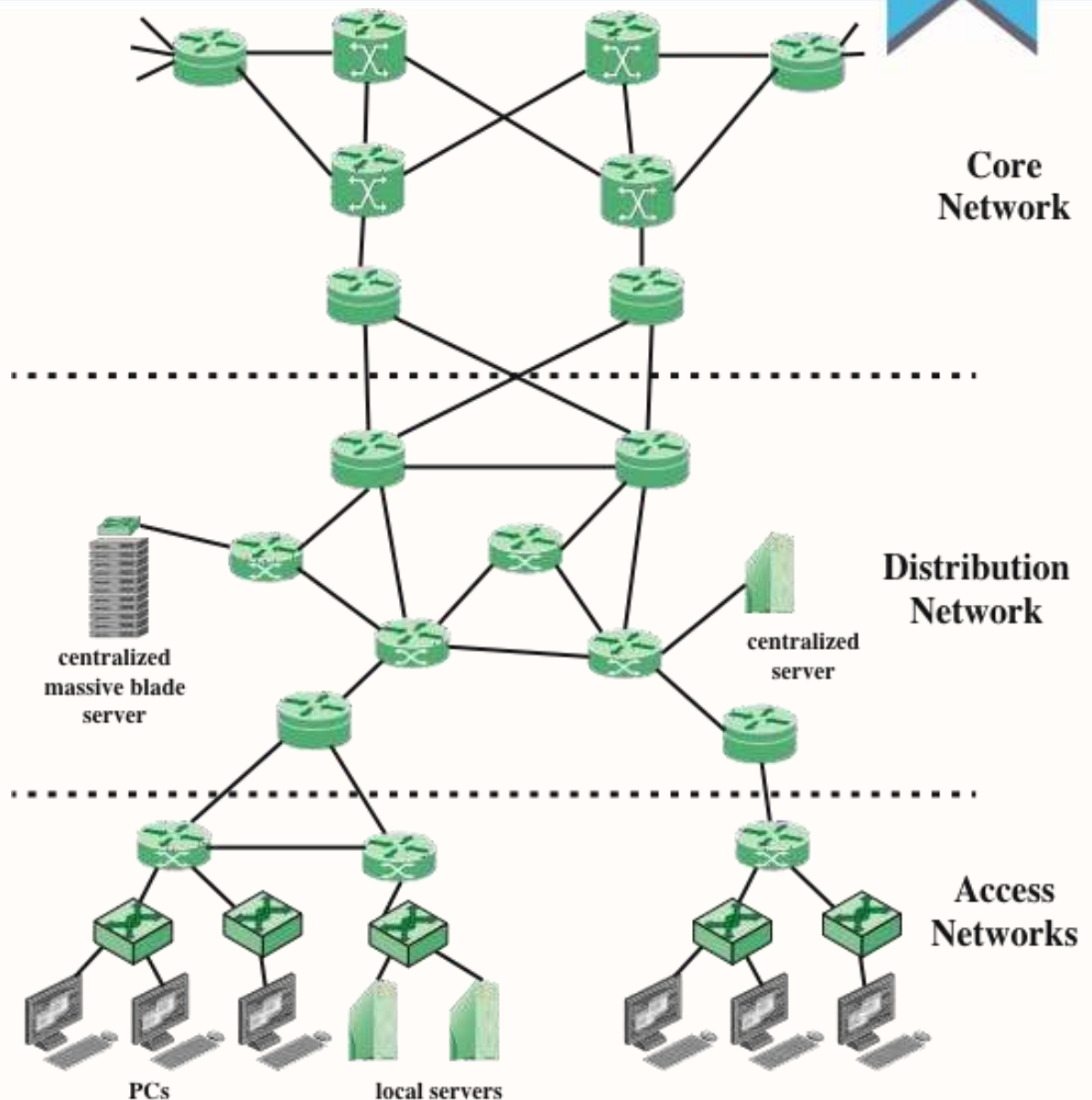


A Global Network Architecture





A Typical Network Hierarchy

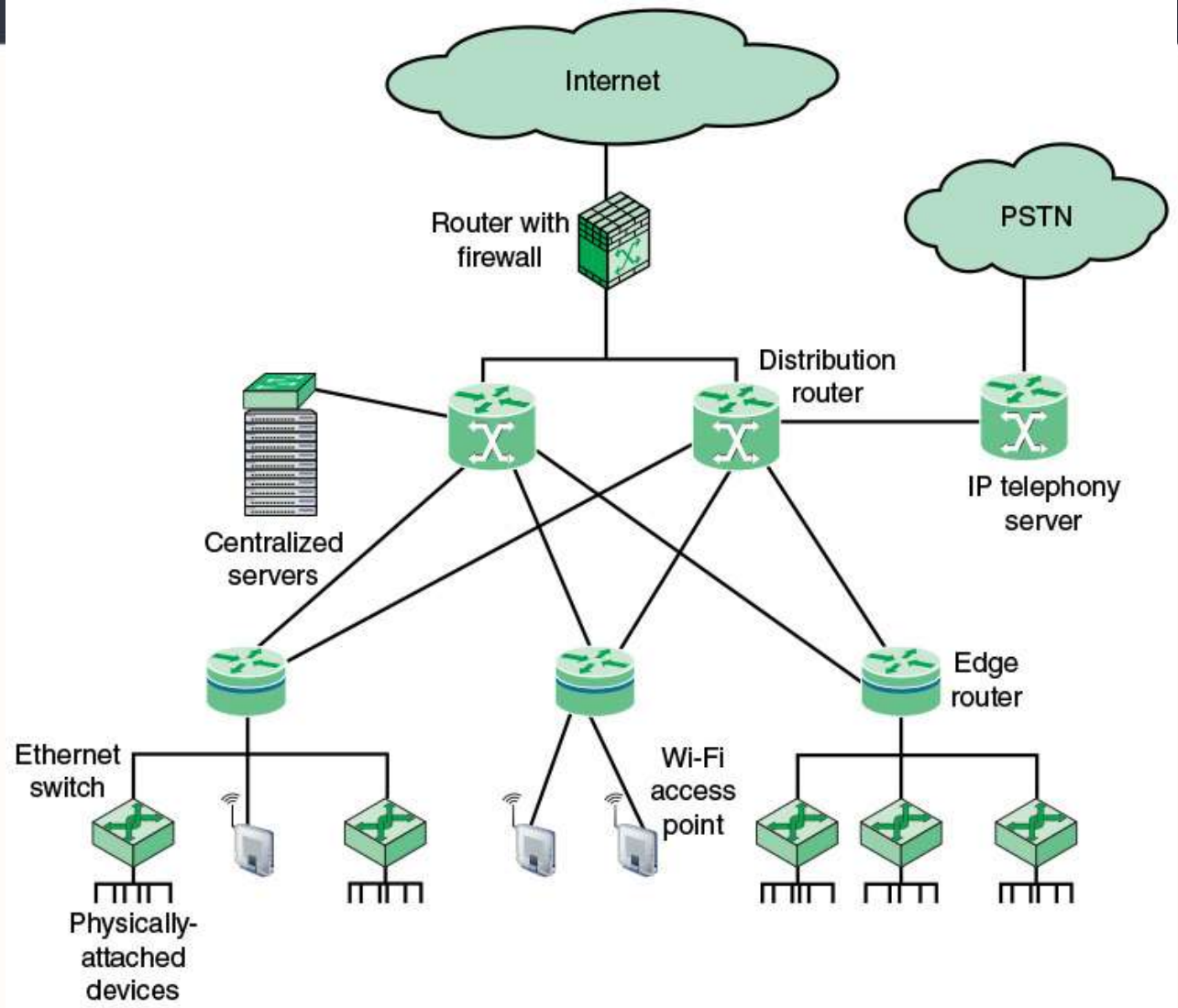




Ethernet

- Predominant wired networking technology
- Supports distances from a few meters to tens of kilometres
- Has evolved to support data rates up to 100 Gbps
- Has become essential for supporting personal computers, workstations, servers, and massive data storage devices in organisations large and small

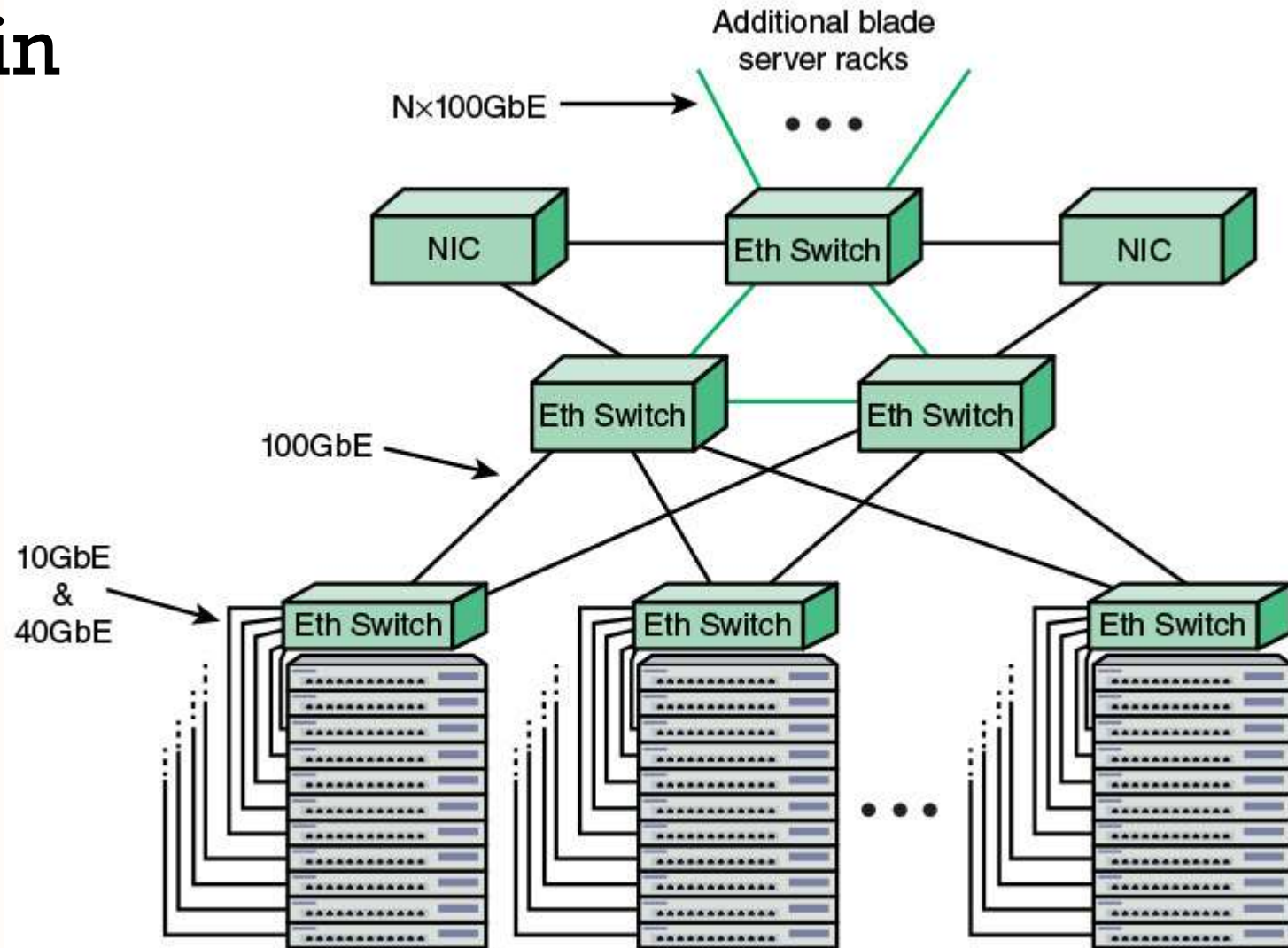
Ethernet in the Office



A Basic Enterprise LAN Architecture



Ethernet in the Data Centre



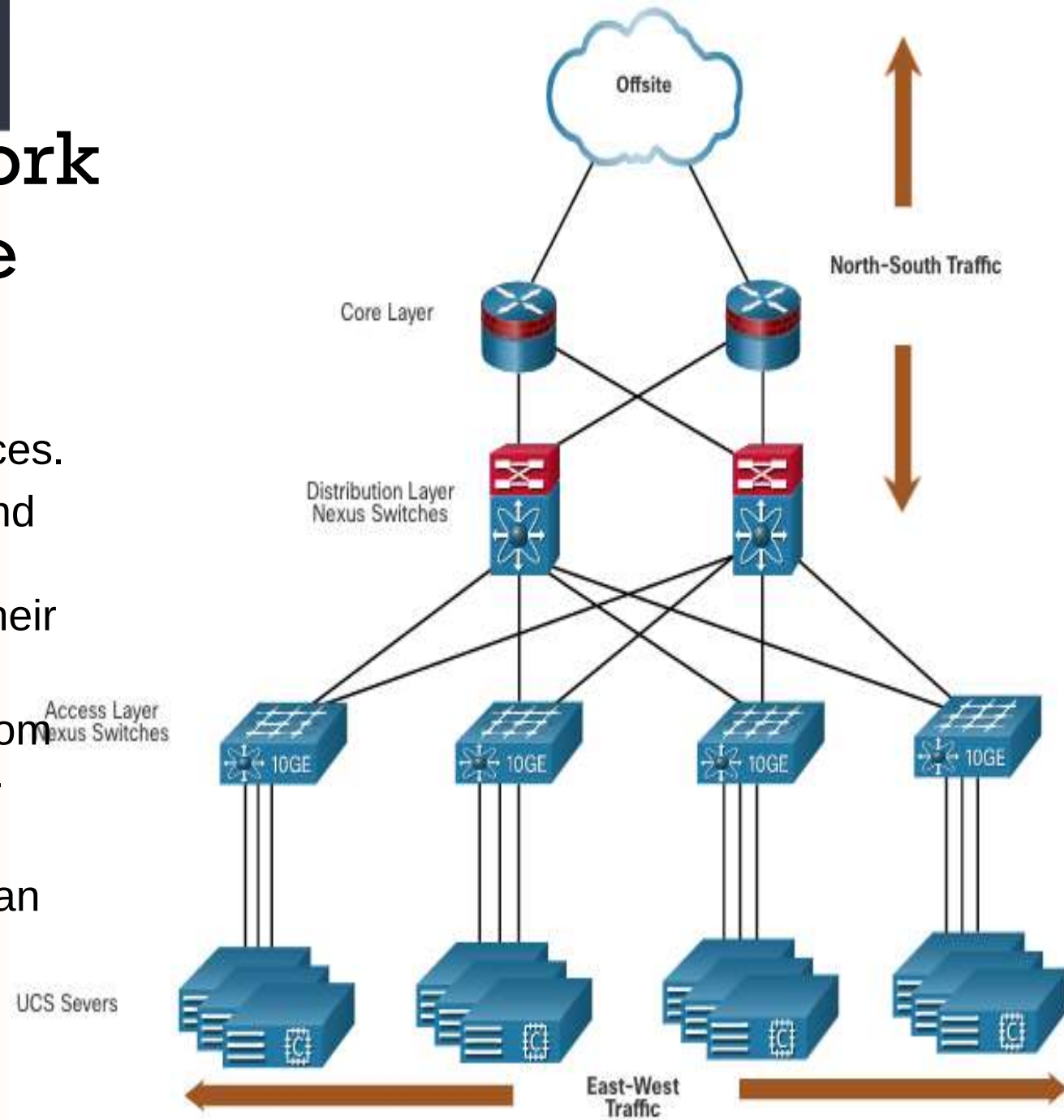


Evolving Network Requirements

- A number of trends are driving network providers and users to reevaluate traditional approaches to network architecture
 - Demand is increasing
 - Supply is increasing
 - Dynamic and complex traffic patterns
 - “horizontal” traffic between servers as well as “vertical” traffic between servers and clients.
 - Elastic and inelastic traffic
 - unpredictable traffic patterns of voice, data, and video
 - Unified communications (UC) strategies

Virtual Network Infrastructure

- Server virtualization hides server resources.
- VMs are movable and the network must change to support their mobility.
- Traffic flows differ from the traditional client-server model.
- Network functions can be virtualized





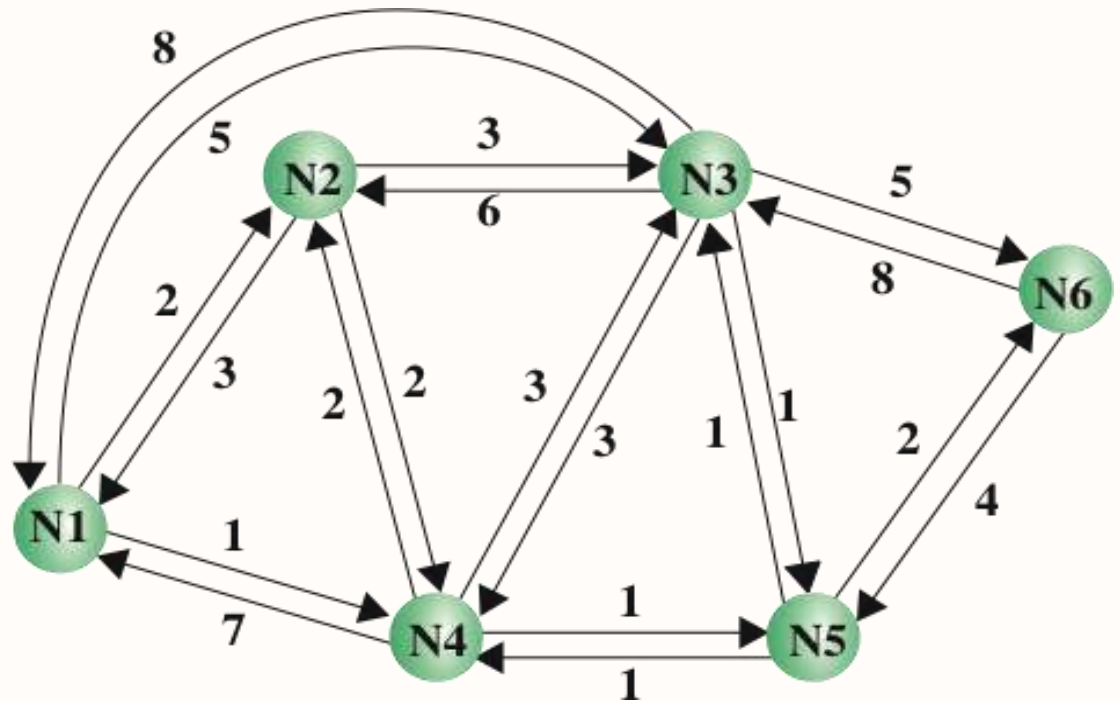
Quality of Experience (QoE)

- A subjective measure of performance as reported by the user; relies on human opinion
- Is important particularly when dealing with multimedia applications and multimedia content delivery
- QoS processes by themselves are not sufficient in that they do not take into account the user's perception of network performance and service quality
- Categories of factors and features that can be included in QoE are:
 - Perceptual
 - Psychological
 - Interactive



Routing Characteristics

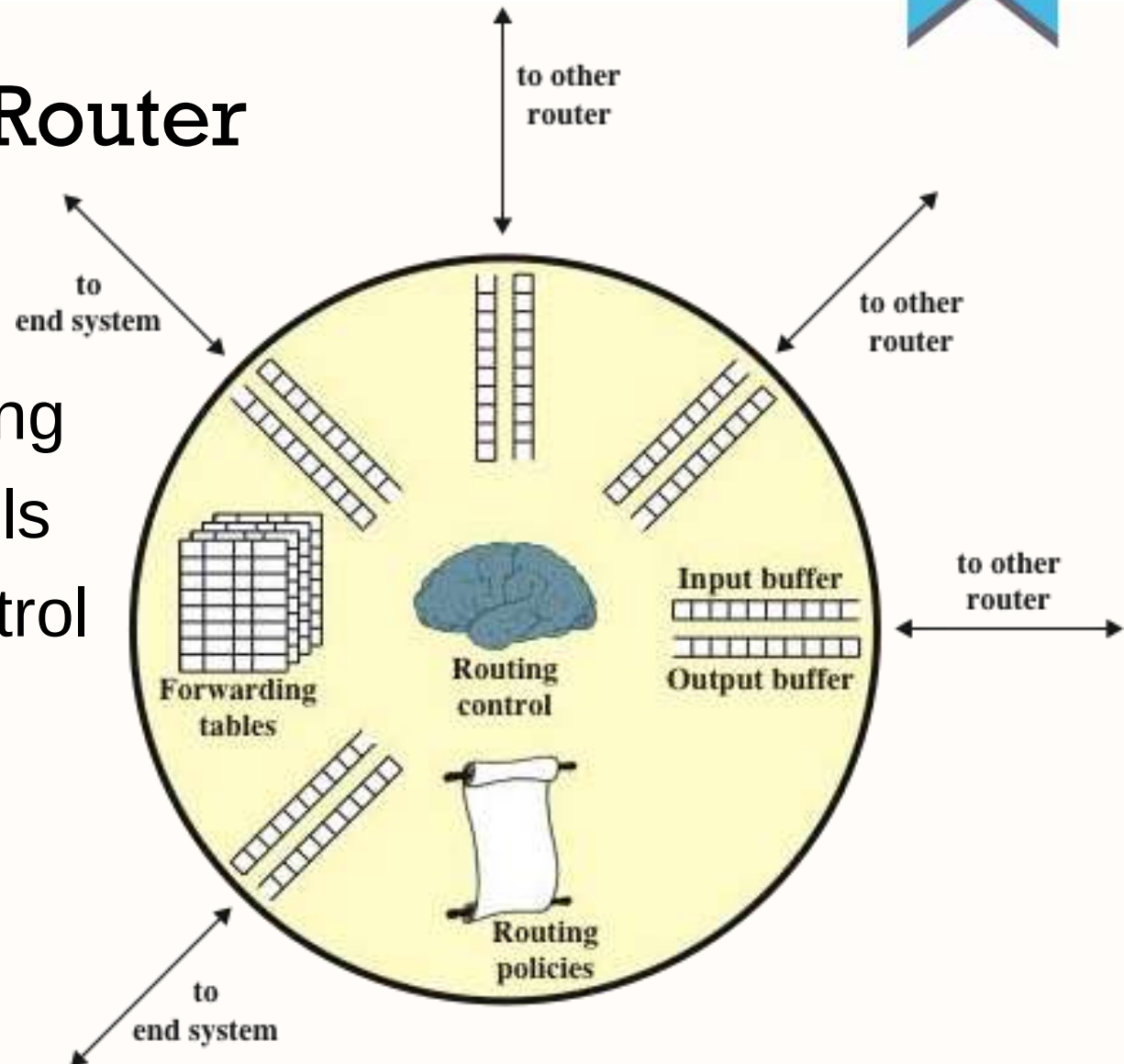
- The primary function of an internet is to accept packets from a source station and deliver them to a destination station
- Selection of a route is generally based on some performance criterion





Elements of a Router

- Packet forwarding
- Routing protocols
- Congestion control





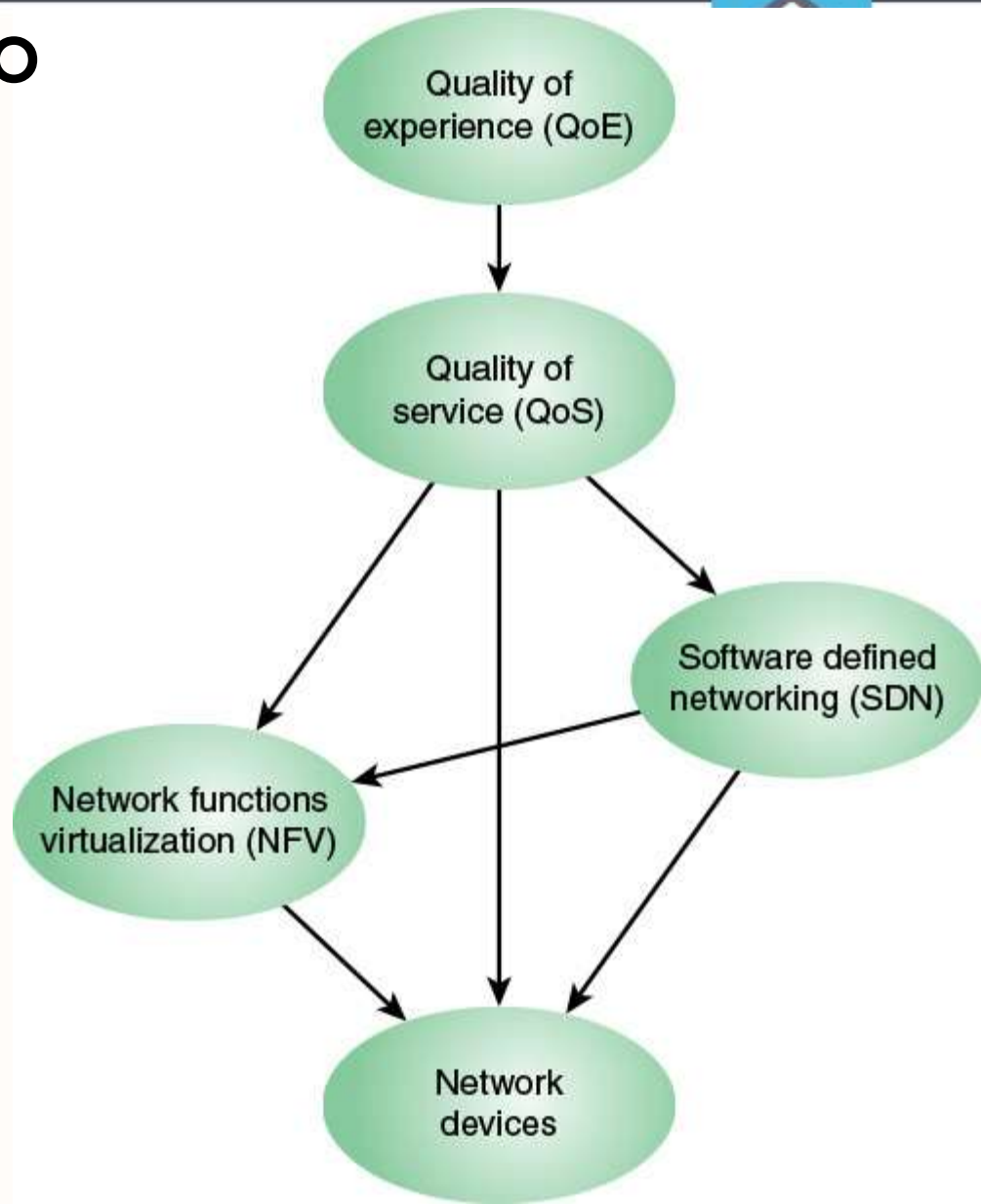
Traditional Network Architectures

- As QoS and QoE requirements imposed on the network are expanded as a result of the variety of applications, the traffic load must be handled in an increasingly sophisticated and agile fashion
- The traditional internetworking approach is based on the TCP/IP protocol architecture; characteristics of this approach are:
 - Two-level end system addressing
 - Routing based on destination
 - Distributed, autonomous control
- The Open Networking Foundation (ONF) cites four general limitations of traditional network architectures:
 - Static, complex architecture
 - Inconsistent policies
 - Inability to scale
 - Vendor dependence



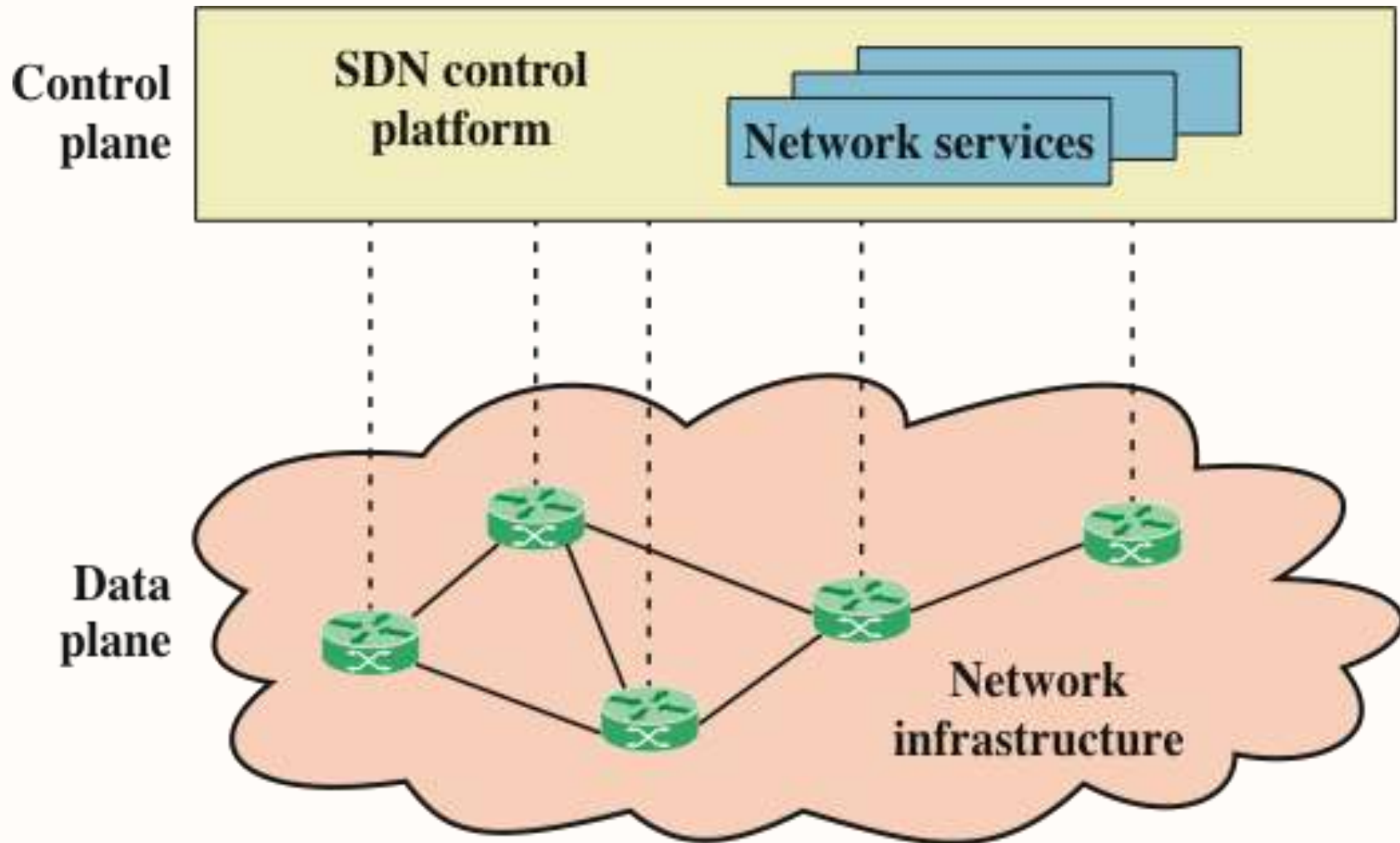
Modern Approach to Networking

- Networks need to be more adaptable and scalable.
- Two key technologies are rapidly being deployed by a variety of network service and application providers
 - Software-Defined Networking (SDN) and
 - Network Functions Virtualization (NFV)





Software Defined Networking



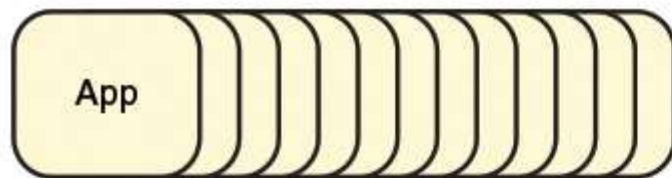


The SDN Approach- Requirements

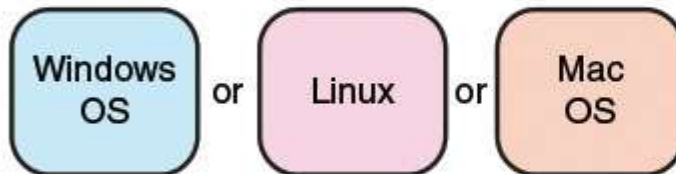
- Adaptability
- Automation
- Maintainability
- Model management
- Mobility
- Integrated security
- On-demand scaling



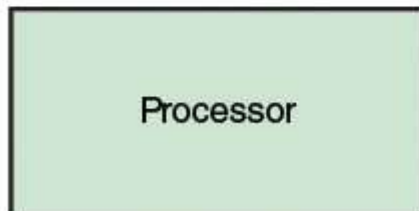
SDN Architecture



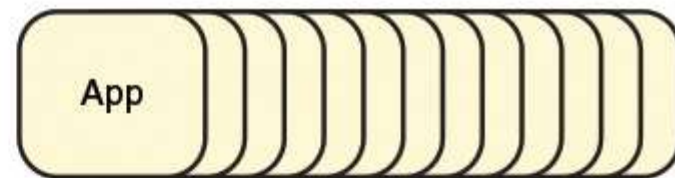
Open Interface



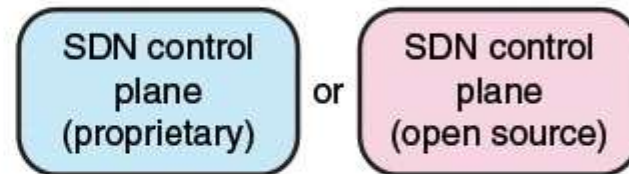
Open Interface



(a) Modern approach
to computing



Open Interface

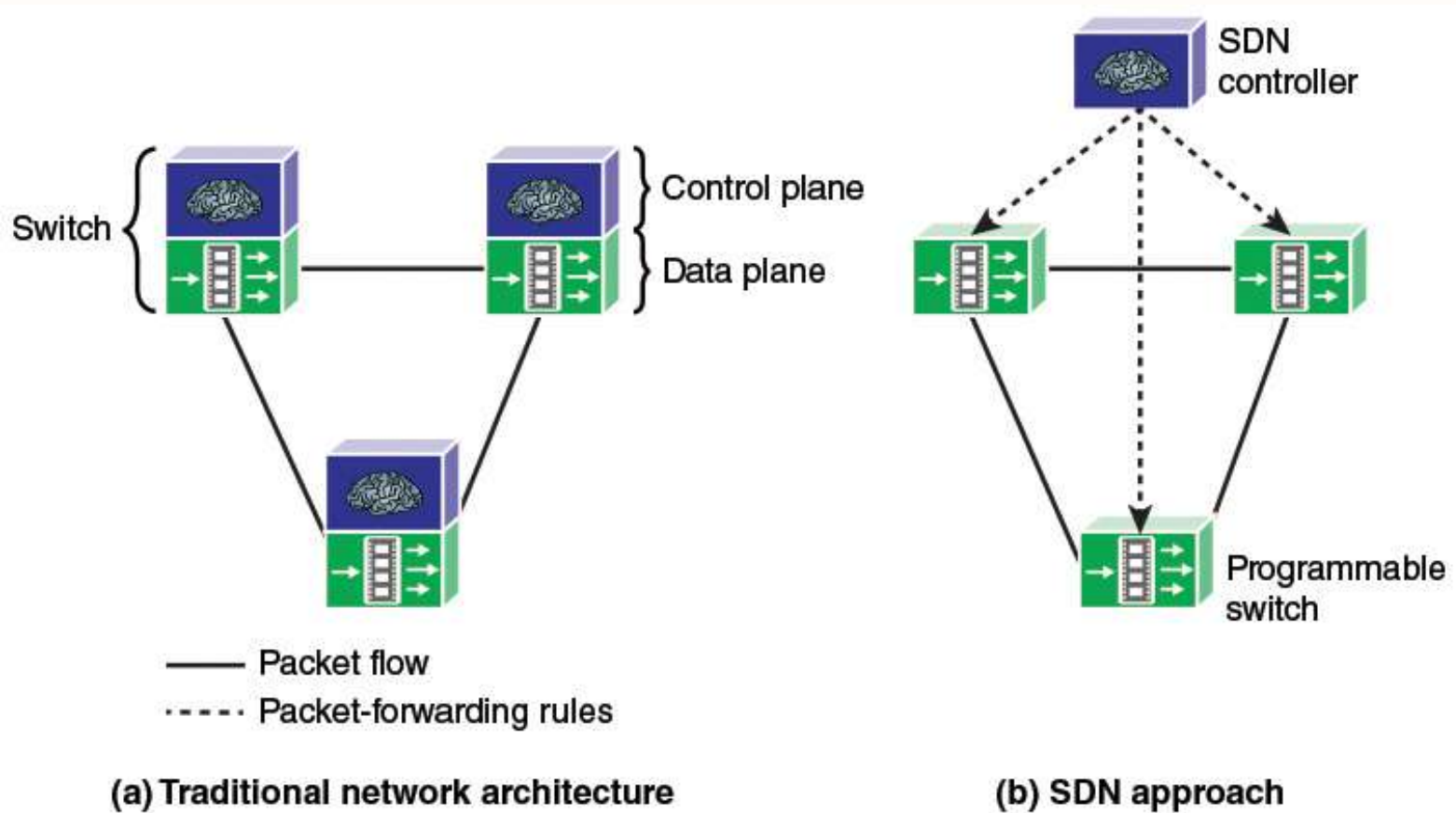


Open Interface

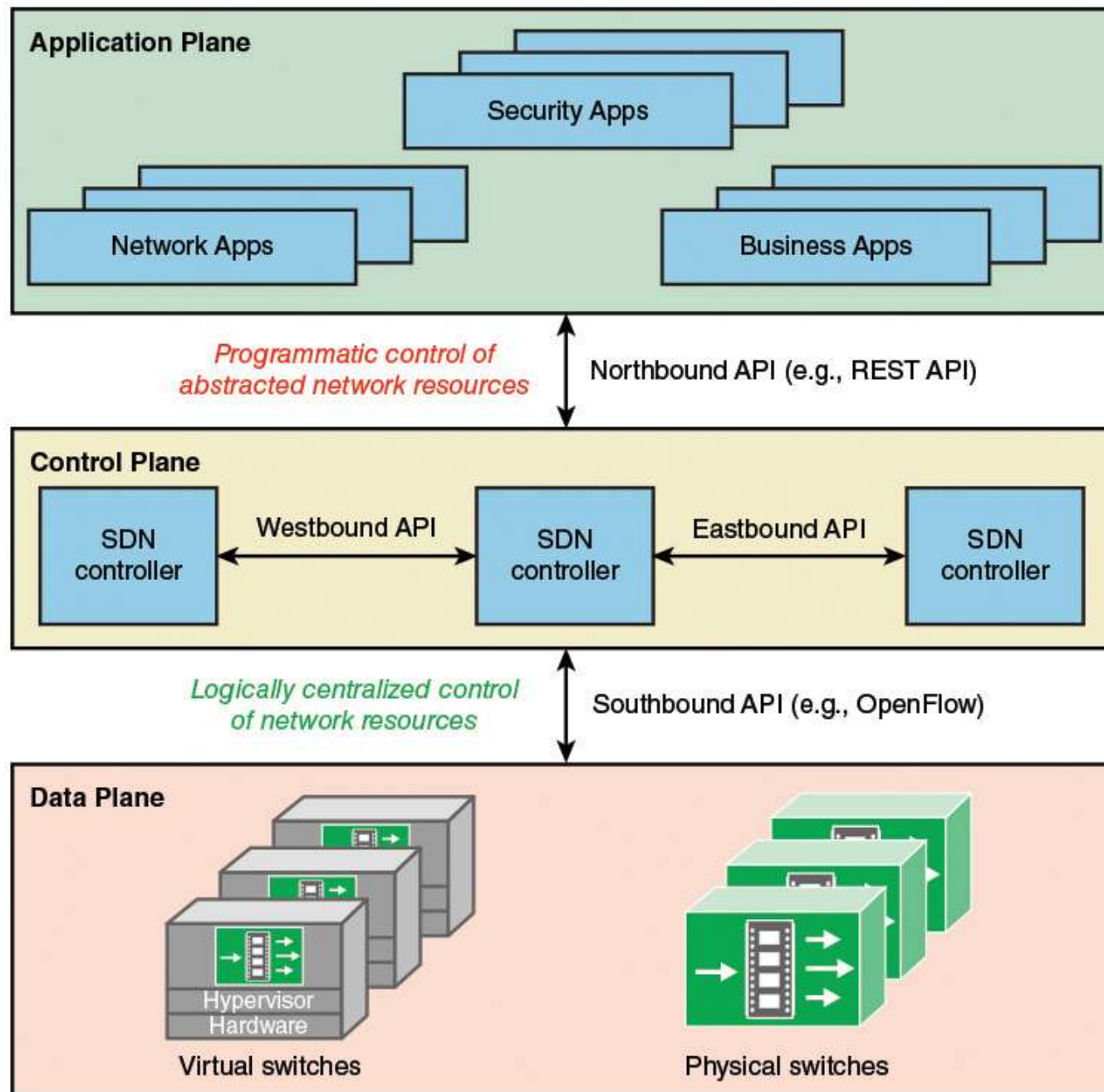


(b) Modern approach
to networking

SDN Architecture



SDN Architecture





Control Plane and Data Plane

A network device contains the following planes:

- **Control plane** - This is typically regarded as the brains of a device. It is used to make forwarding decisions. The control plane contains Layer 2 and Layer 3 route forwarding mechanisms, such as routing protocol neighbor tables and topology tables, IPv4 and IPv6 routing tables, STP, and the ARP table. Information sent to the control plane is processed by the CPU.
- **Data plane** - Also called the forwarding plane, this plane is typically the switch fabric connecting the various network ports on a device. The data plane of each device is used to forward traffic flows. Routers and switches use information from the control plane to forward incoming traffic out the appropriate egress interface. Information in the data plane is typically processed by a special data plane processor without the CPU getting involved.

Example



Cisco ACI | Architecture

Any Cloud, Any Application, Any Platform, Anywhere





Organization	Mission	SDN-related effort
Open Networking Foundation (ONF)	An industry consortium dedicated to the promotion and adoption of SDN through open standards development.	OpenFlow
Internet Engineering Task Force (IETF)	The Internet's technical standards body. Produces RFCs and Internet Standards.	Interface to Routing Systems (I2RS) Service Function Chaining
European Telecommunications Standards Institute (ETSI)	An EU-sponsored standards organization that produces globally-applicable standards for information and communications technologies.	NFV Architecture
OpenDaylight	A collaborative project under the auspices of the Linux Foundation	OpenDaylight
International Telecommunication Union – Telecommunication Standardization Sector (ITU-T)	United Nations agency that produces Recommendations with a view to standardizing telecommunications on a worldwide basis.	SDN functional requirements and architecture
Internet Research Task Force (IRTF) Software Defined Networking Research Group (SDNRG)	Research group within IRTF. Produces SDN-related RFCs.	SDN architecture
Broadband Forum (BBF)	Industry consortium developing broadband packet networking specifications.	Requirements and framework for SDN in telecommunications broadband networks
Metro Ethernet Forum (MEF)	Industry consortium that promotes the use of Ethernet for metropolitan and wide area applications.	Defining APIs for service orchestration over SDN and NFV
IEEE 802	An IEEE committee responsible for developing standards for local area networks (LANs)	Standardize SDN capabilities on access networks.
Optical Internetworking Forum (OIF)	Industry consortium promoting development and deployment of interoperable networking solutions and services for optical networking products	Requirements on Transport Networks in SDN Architectures

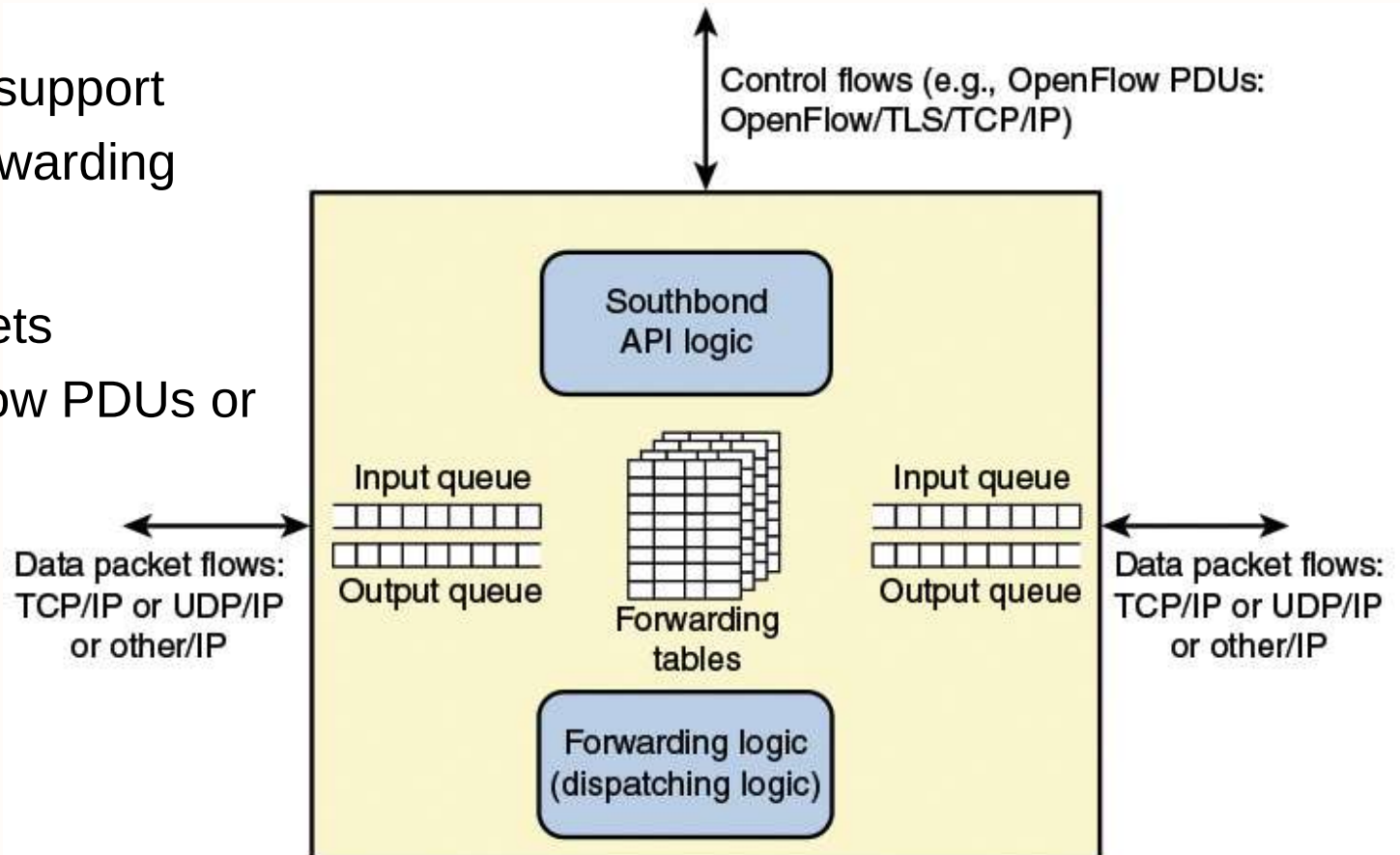
Organization	Mission	SDN-related effort
Open Data Center Alliance (ODCA)	Consortium of leading IT organizations developing interoperable solutions and services for cloud computing.	SDN Usage Model
Alliance for Telecommunications Industry Solutions (ATIS)	A standards organization that develops standards for the unified communications (UC) industry.	Operational Opportunities and Challenges of SDN/NFV Programmable Infrastructure
Open Platform for NFV (OPNFV)	An open source project focused on accelerating the evolution of NFV.	NFV infrastructure

SDN and NFV Standards Activities



Data Plane Network Device

- Functions
 - Control support
 - Data forwarding
- Protocols
 - IP packets
 - OpenFlow PDUs or similar





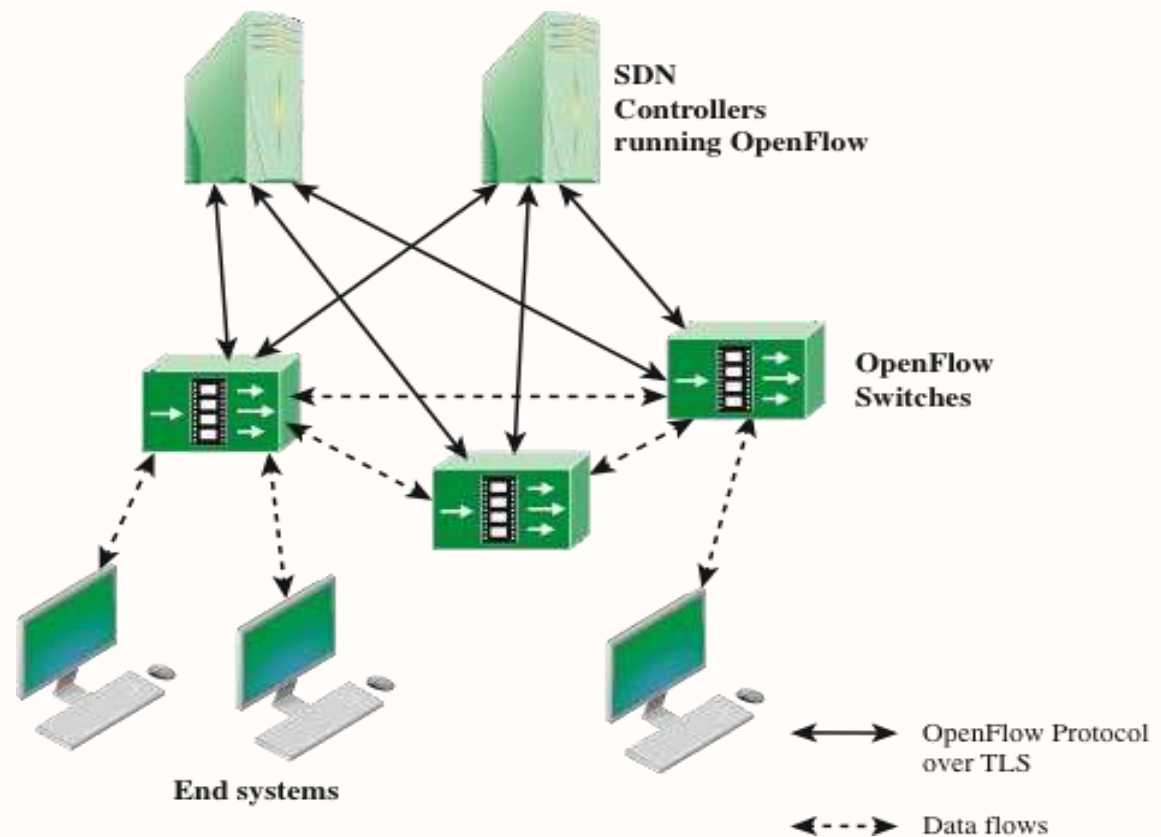
OpenFlow Logical Network Device

- SDN practical implementation requires:
 - A common logical architecture in all switches, routers, and other network devices to be managed by an SDN controller.
 - May be implemented in different ways as long as the SDN controller sees a uniform logical switch functionality.
 - A standard, secure protocol is needed between the SDN controller and the network device.
- These requirements are addressed by OpenFlow
 - OpenFlow is both a protocol between SDN controllers and network devices and a specification of the logical structure of the network switch functionality.
 - OpenFlow is defined in the OpenFlow Switch Specification, published by the Open Networking Foundation (ONF).



OpenFlow Switch Context

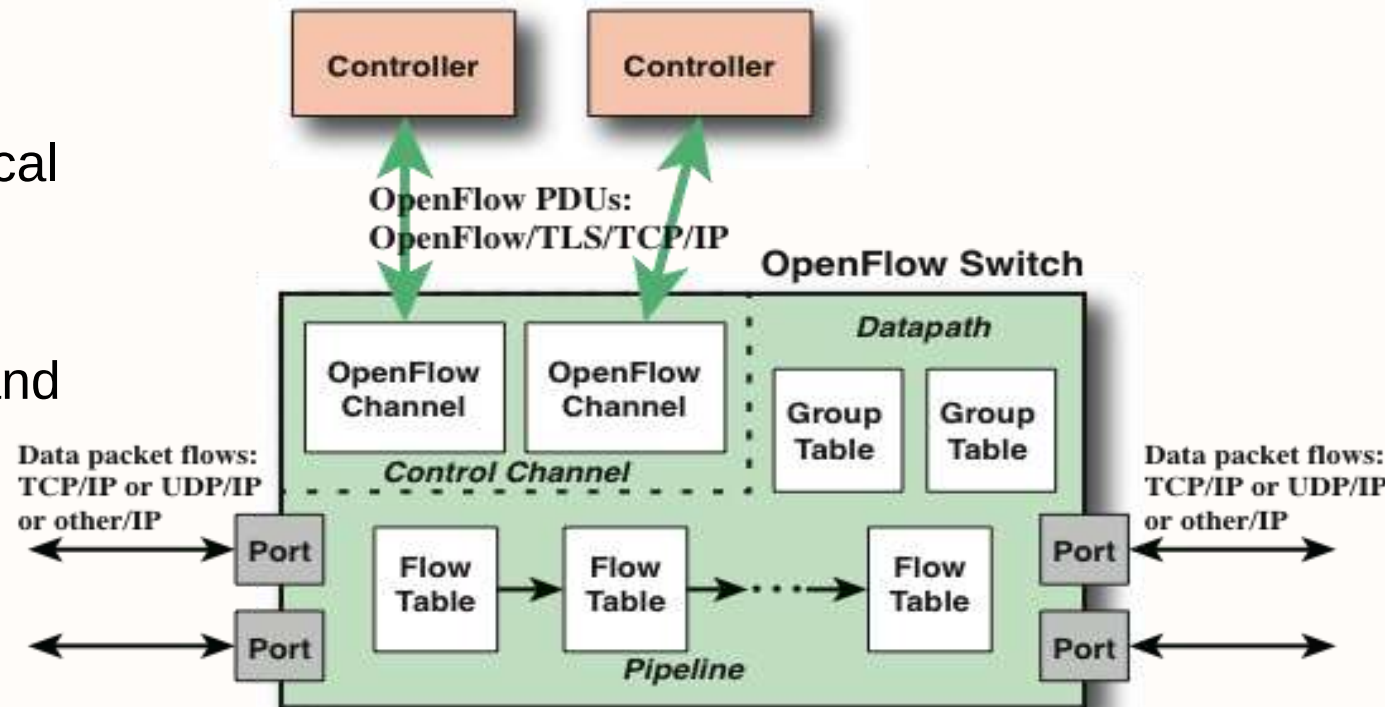
- Main elements of an OpenFlow environment, consisting of SDN controllers that include OpenFlow software, OpenFlow switches, and end systems





Main Components of an OpenFlow Switch

- Ports
 - Physical, logical and reserved
- Tables
 - Flow, group and meter



A flow is a sequence of packets traversing a network that share a set of header field values such as IP addresses or VLAN id



Flow Table Structure

Match fields	Priority	Counters	Instructions	Timeouts	Cookie	Flags
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(a) Flow Table Entry Fields

Ingr port	Egr port	Ethr SA	Ethr DA	Ethr Type	IP port	IPv4 SA	IPv4 DA	IPv6 SA	IPv6 DA	TCP Src	TCP Dest	UDP Src	UDP Dest
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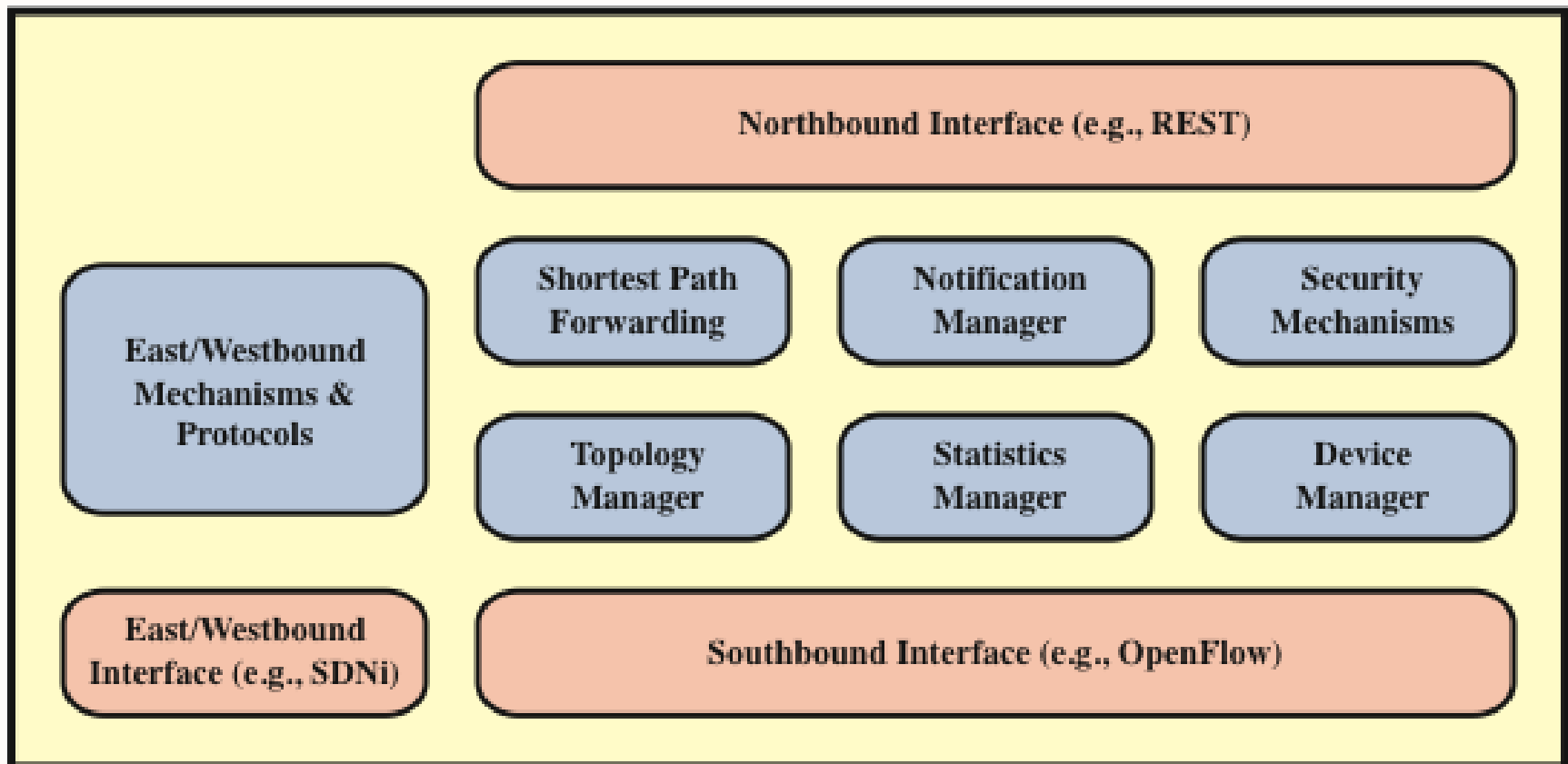
(b) Flow Table Match Fields (required fields)

Group Identifier	Group Type	Counters	Action Buckets
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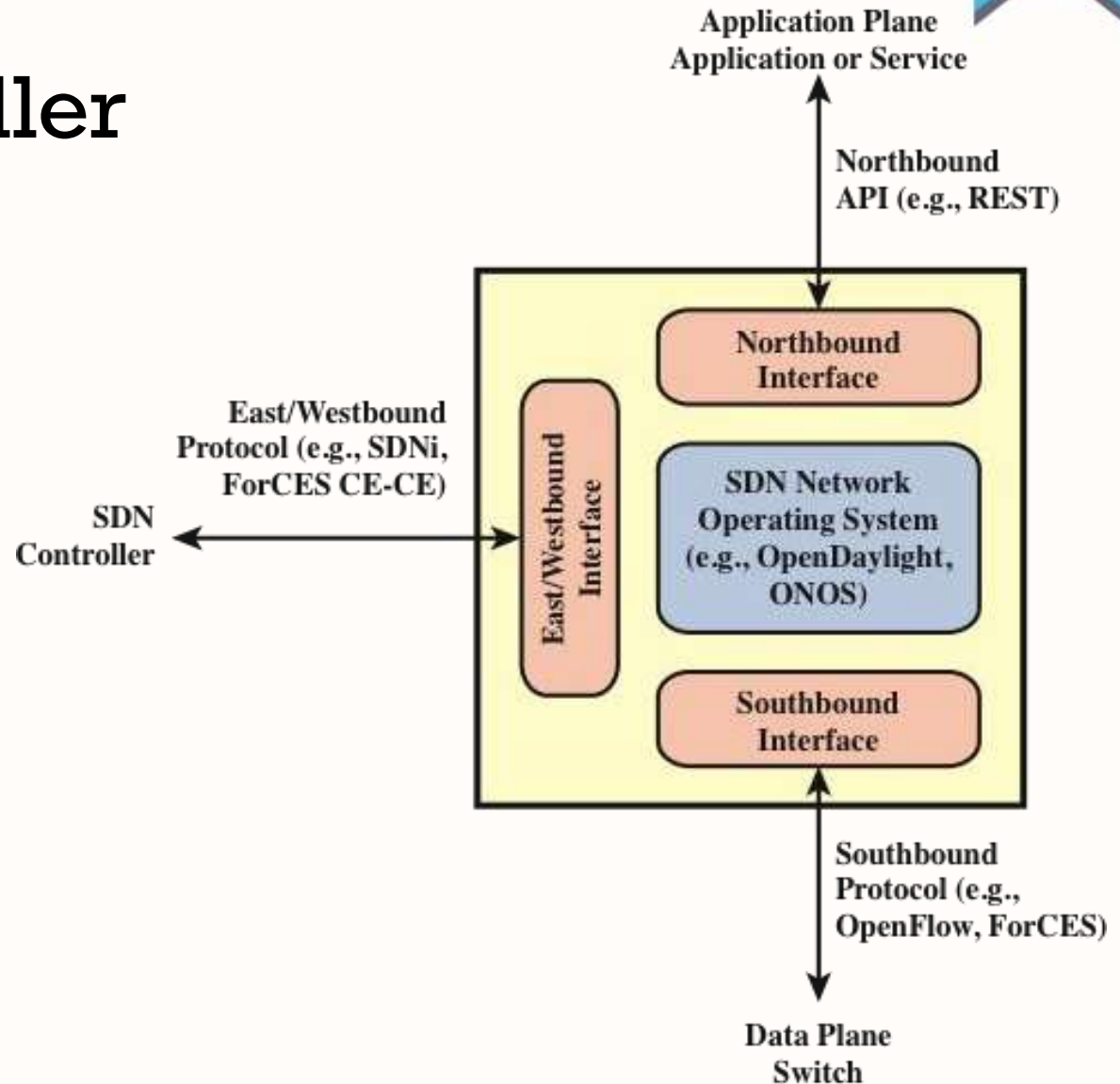
(c) GroupTable Entry Fields



SDN Control Plane Functions and Interfaces



SDN Controller Interfaces



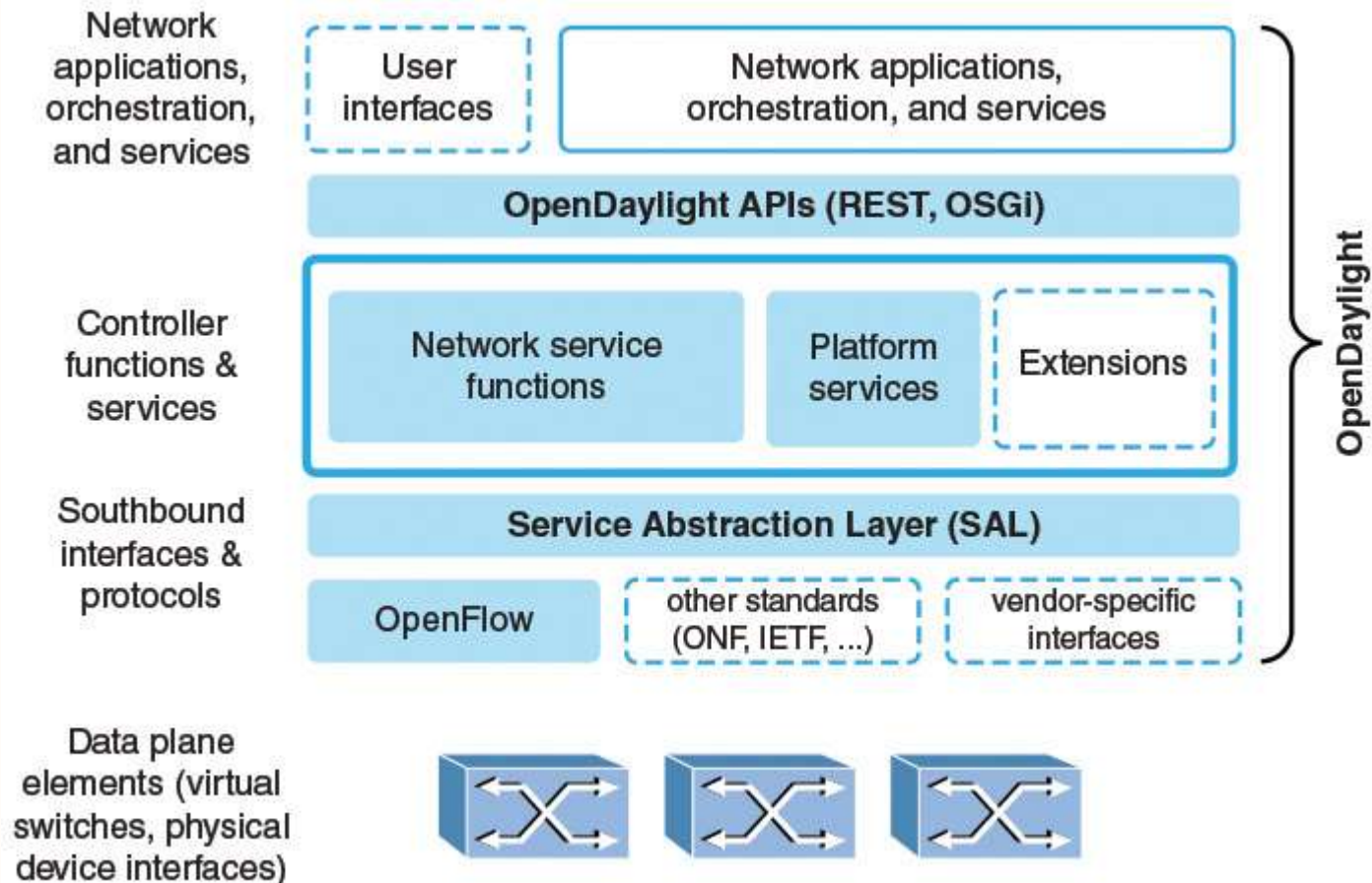


Routing

- Traditionally the routing function is distributed among the routers in a network
- In an SDN controlled network, it makes sense to centralize the routing function within the SDN controller
- The controller can develop a consistent view of the network state for calculating shortest paths and can implement application aware routing policies
- The data plane switches are relieved of the processing and storage burden associated with routing, leading to improved performance
- The centralized routing application performs two distinct functions:
 - Link discovery
 - Topology manager

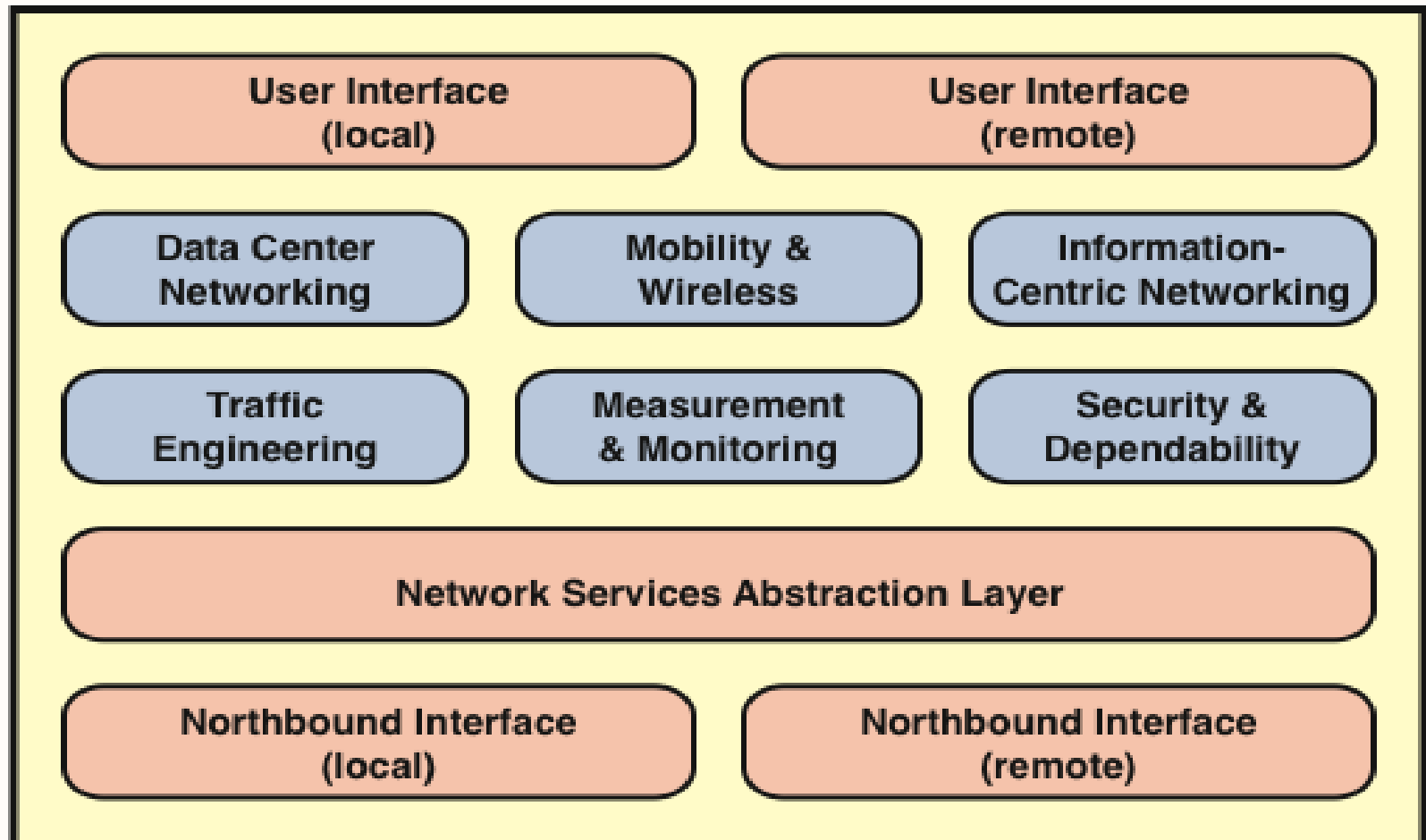


OpenDaylight SDN controller





SDN Application Plane





Network Functions Virtualization (NFV)

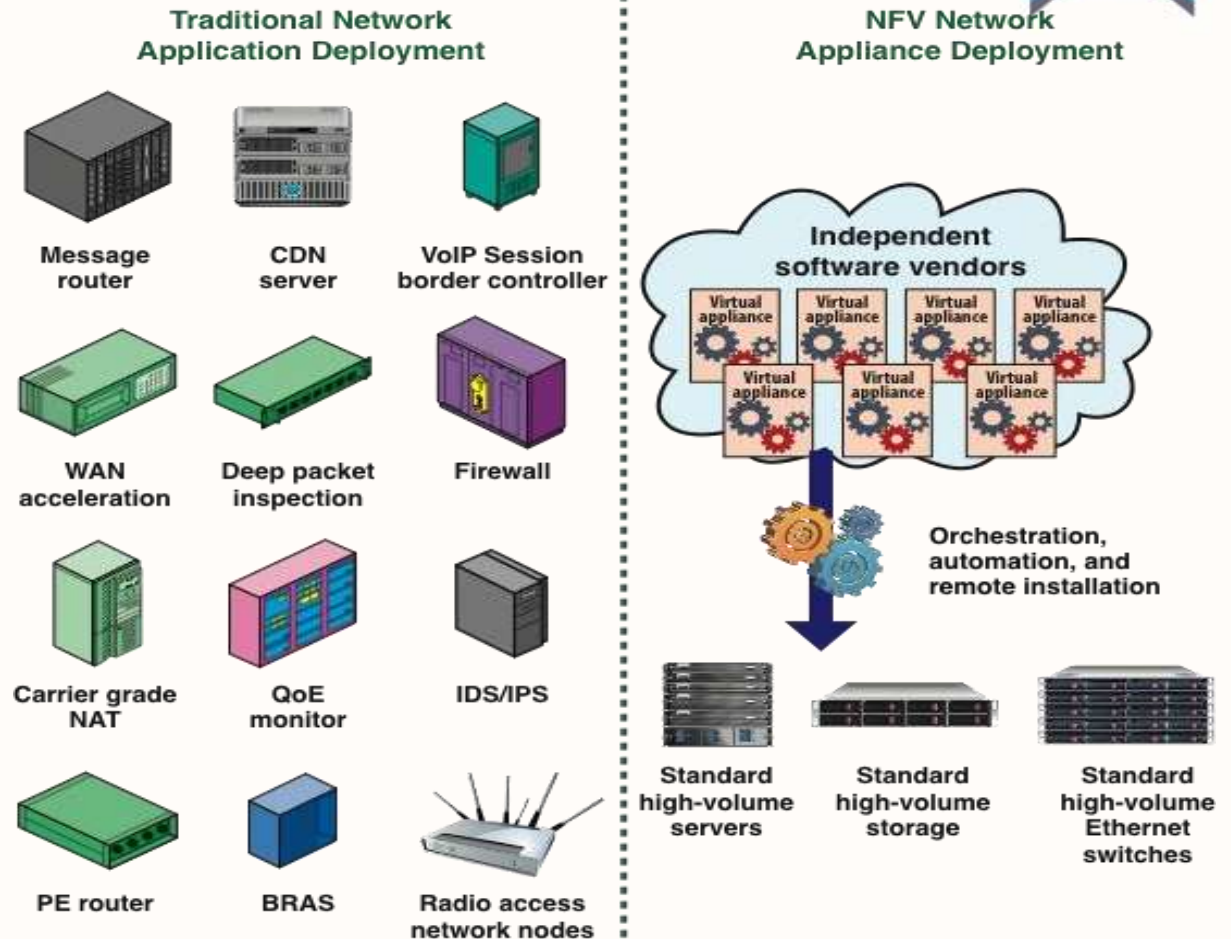
- Motivation
 - New network services may require additional different types of hardware appliances
 - New hardware means additional capital expenditures.
 - Skills necessary to design, integrate, and operate increasingly complex hardware-based appliances
 - Hardware-based appliances rapidly reach end of life
 - The need for an increasing variety of hardware platforms inhibits the introduction of new revenue-earning network services
- The NFV approach moves away from dependence on a variety of hardware platforms to the use of a small number of standardized platform types, with virtualization techniques used to provide the needed network functionality.



NFV

- NFV decouples network functions from proprietary hardware appliances so that they can run in software on VMs
 - Network function devices: Such as switches, routers, network access points, customer premises equipment (CPE), and deep packet inspectors (for deep packet inspection).
 - Network-related compute devices: Such as firewalls, intrusion detection systems, and network management systems.
 - Network-attached storage: File and database servers attached to the network.
- Builds on standard VM technologies, extending their use into the networking domain

Vision for NFV



CDN = content delivery network
WAN = wide area network
NAT = network address translation
QoE = quality of experience
VoIP = voice over Internet Protocol

IDS = intrusion detection system
IPS = intrusion prevention system
PE = provider edge router
BRAS = broadband remote access server



Questions?